FYISC - ANGLE AND IT'S MEASUREMENT

Exercise - 1

1. (a) Draw diagrams for the following angles:

(i) $-135°$ (ii) $740°$.

In which quadrant do they lie?

Ans. (i) Third quadrant (ii) First quadrant

(b) Find another positive angle whose initial and final sides are same as that of $-135°$ and indicate on the same diagram.

Ans. $225°$

2. If θ lies in second quadrant, in which quadrant the following will lie?

(i) $\frac{\theta}{2}$ (ii) 2θ

(iii) $-\theta$.

Ans. (i) First quadrant (ii) Third or fourth quadrant

(iii) Third quadrant

3. Express the following angles in radian measure:

(i) 240^0 (ii) -315^0 (iii) $570°$.

Ans. (i) $\frac{4\pi}{3}$ (ii) $-\frac{7\pi}{4}$ (iii) $\frac{19\pi}{6}$

4. Express the following angles in degree measure:

(i) $\frac{5\pi}{3}$ (ii) $\frac{13\pi}{4}$ (iii) $-\frac{24\pi}{5}$.

Ans. (i) 300^0 (ii) 585^0 (iii) -864^0

5. Express the following angles in radian measure :

(i) 35^0 (ii) 520^0 (iii) $40°20'$ (iv) $-37°30'$

Ans. (i) $\frac{7\pi}{36}$ (ii) $\frac{26\pi}{9}$ (iii) $\frac{121\pi}{540}$ (iv) $-\frac{5\pi}{24}$

6. Find the degree measures corresponding to the following radian measures:

(i) 6 (ii) $\frac{3}{4}$ (iii) -3.

Ans. (i) $343°38'11''$ (ii) $42°57'16''$ (iii) $-171°49'5''$

7. A wheel makes 360 revolutions in a minute. Through how many radians does it turn in one second ?

Ans. 12π

8. Find the angle in radians through which a pendulum swings if its length is 75 cm and the tip describes an arc of length :

(i) 10 cm (ii) 15 cm.

Ans. (i) $\frac{2}{15}$ (ii) $\frac{1}{5}$

9. Find the radius of the circle in which a central angle of 45^0 makes an arc of length 187 cm. $\left(\text{use } \pi = \frac{22}{7}\right)$.

Ans. 238 cm

10. Find the length of an arc of a circle of diameter 20 cm which subtends an angle of 45^0 at the centre.

Ans. $\frac{5\pi}{2}$ cm

11. An engine is travelling along a circular railway track of radius 1500 metres with a speed of 60 km/h. Find the angle in degrees turned by the engine in 10 seconds.

Ans. $\left(\frac{20}{\pi}\right)^\circ$

12. If the arcs of the same length in two circles subtend angles of 65° and 110° at their respective centres, find the ratio of their radii.

Ans. 22 : 13

13. Large hand of a clock is 21 cm long. How much distance does its extremity move in 20 minutes?

Ans. 44 cm

14. Find the angles in degrees through which a pendulum swings if its length is 50cm and the tip describes an arc of length:

(i) 10cm (ii) 16 cm (iii) 26 cm $\left(\text{use } \pi = \frac{22}{7}\right)$.

Ans. (i) 11° 27'16" (ii) 18° 19'38" (iii) 29° 46'55"

15. Find the length of an arc of a circle of radius 75 cm that spans a central angle of measure 126°. Take $\pi = 3.1416$.

Ans. 164 . 934 cm

16. Find the angle in radians between the hands of a clock at 3.30 a.m.

Ans. $\frac{5}{12}\pi$

17. The circular measures of two angles of a triangle are $\frac{1}{2}$ and $\frac{1}{3}$. Find the third angle in degree measure. Take $\pi = \frac{22}{7}$.

Ans. $132°16'22''$

18. The difference between two acute angles of a right angled triangle is $\frac{\pi}{5}$ in radian measure. Find these angles in degrees.

Ans. $63°, 27°$

19. The angles of a triangle are in A.P. and the greatest angle is double the least. Find all the angles in circular measure.

Ans. $\frac{2\pi}{9}, \frac{\pi}{3}, \frac{4\pi}{9}$

20. Estimate the diameter of the sun supposing that it subtends an angle of $32'$ at the eye of an observer. Given that the distance of the sun is 91×10^6 km. Take $\pi = \frac{22}{7}$.

Ans. 847407.4 km

Exercise - 2

1. Convert the following into radian measures :

(i) $25°$ (ii) $-47° 30'$ (iii) $5° 37'30''$

Ans. (i) $\frac{5\pi}{36}$ radians. (ii) $-\frac{19\pi}{72}$ radians. (iii) $\frac{\pi}{32}$ radians.

2. Convert the following radian measures into degree measures $\left(\text{use } \pi = \frac{22}{7}\right)$.

(i) $\frac{11}{16}$ (ii) -4 (iii) $\frac{7\pi}{6}$.

Ans. (i) $39° 22'30''$ (ii) $-229° 5'27''$ (iii) 210^0

3. Express in radians the fourth angle of a quadrilateral which has three angles $46° 30'10''$, $75° 44'45''$ and $123° 9'35''$. Take $\pi = \frac{22}{7}$

Ans. 2 radians

4. Find the angle in radian through which a pendulum swings if its length is 75 cm and the tip describes an arc of length 21 cm.

Ans. $\frac{7}{25}$

5. Find the radius of the circle in which a central angle of 60° intercepts an arc of 37.4 cm length. $\left(\text{use } \pi = \frac{22}{7}\right)$.

Ans. 35.7 cm

6. A circular wire of radius 3 cm is cut and bent so as to lie along a circumference of a hoop whose radius is 48 cm. Find the angle in degrees which is subtended at the centre of the hoop.

Ans. $22\frac{1}{2}^{\circ}$.

7. The large hand of a big clock is 35 cm long. How many cm does the tip move in 9 minutes ?

Ans. 33 cm

8. Find the angle in radians between the hands of a clock at 7.20 p.m.

Ans. $\frac{5\pi}{9}$ radians

9. A wheel of a motor is rotating at 1200 r.p.m. If the radius of the wheel is 35 cm, what liner distance does a point of its rim traverse in 30 seconds ?

Ans. 1.32 km.

10. If the angles of a triangle are in the ratio 3 : 4: 5, find the smallest angle in degrees and the greatest angle in radians.

Ans. $\frac{5\pi}{12}$ radians.

11. The angles of a triangle are in A.P. and the number of degrees in the least to the number of radians in the greatest is 60 : π. Find the angles in degrees and radians.

Ans. $\frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}$.

12. Taking the moon's distance from the earth as 360000 km and the angle subtended by the moon at any point O on the earth as half a degree, estimate the diameter of the moon. (Use π = 3.1416)

Ans. 3141.6 km.

13. The perimeter of a certain sector of a circle is equal to the length of the arc of a semicircle having the same radius. Find the angle of the sector in degrees. Take $\pi = \frac{22}{7}$.

Ans. 65° 27'18''

14. A sector of a circle of radius r metres is bounded by an arc AB and by two radii OA and OB, at an angle β radians. Given that the perimeter of the sector is 18 m and that the area of the sector is 8 m^2, calculate the numerical values of r and β.

Ans. $r = 8, \beta = \frac{1}{4}$.

15. Express in radians and also in degrees the angle of a regular polygon of

(i) 40 sides, (ii) n sides.

Ans. $\frac{19}{20}\pi, 171^\circ, \left(\frac{n-2}{n}\right)\pi$ rad, $\left(\frac{n-2}{n}\right)180^\circ$.

16. In a circle of diameter 40 cm, the length of a chord is 20 cm. find the length of minor and major arc of the chord.

Ans. $\frac{20\pi}{3}$ cm, $\frac{100\pi}{3}$ cm

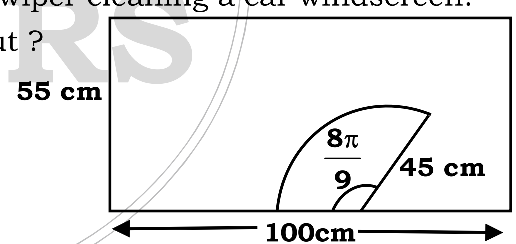
17. Find the area of sector of a circle, radius 5 m bounded by an arc of length 8 m.

Ans. 20 m^2

18. The diagram in fig., shows a windscreen wiper cleaning a car windscreen.

(i) What is the length of the arc swept out ?

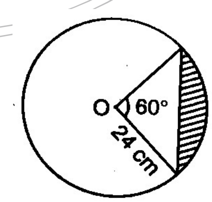
(ii) What area of the windscreen is not cleaned ? (π = 3.14)



Ans. (i) 126 cm

(ii) 2674 cm^2

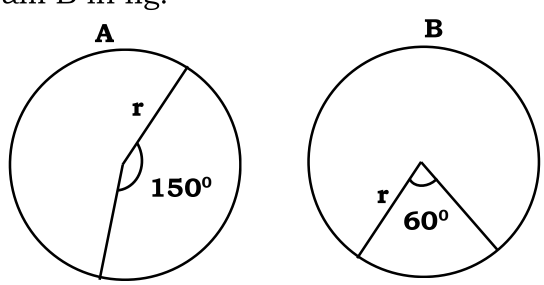
19. Find the area of the shaded segment.



Ans. 52.3 cm^2

20. What is the ration of the areas of the major sector in diagram A to the minor sector in a diagram B in fig.

Ans. 7 : 2



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